

SENIOR SAAS PRODUCT ARCHITECT + MERN CODE AUDITOR

Lead Management Module Audit Report

CodeBegun LMS — Comprehensive Codebase Audit

ORGANIZATION	DATE	STACK	SECTIONS
CodeBegun	April 26, 2026	MERN + WhatsApp + Meta Ads + OpenAI	10 Sections

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10-section comprehensive audit covering schema, APIs, UI, AI readiness, and implementation roadmap

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Audit Score Summary

27 Features Available 	14 Partially Working 	18 Features Missing 	5 Critical / P0 Bugs 
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Lead Model / Schema Audit

server/src/models/Lead.ts — All fields, sub-documents, and indexes

Feature / Field	Status	Notes
Basic fields (name, phone, email)	✓ Available	phone required, email optional
Course interest	✓ Available	Array of strings, multiple courses supported
Source tracking	✓ Available	source string + full sourceDetails sub-doc (platform, campaignId, adSetId, adId, formId, landingPage)
UTM parameters	✓ Available	Full utmParams sub-doc (source, medium, campaign, content, term)
Stage tracking	✓ Available	stageId → ref to LeadStage collection
Priority (Hot/Warm/Cold)	✓ Available	priority enum + score number
Lead assignment	✓ Available	assignedTo (current) + full assignment sub-doc with previousAssignees history
Follow-up date	✓ Available	nextFollowUp Date field
Notes	✓ Available	notes string
Activity / Timeline log	✓ Available	activities[] — note, call, email, whatsapp, status_change, assignment, created
Call recording URL	✓ Available	activities[].recordingUrl
Call outcome	✓ Available	not_answered, busy, rejected, connected, not_connected
Call duration	✓ Available	activities[].callDuration (seconds)
WhatsApp tracking	✓ Available	whatsappStatus, whatsappReplied, whatsappEngagement sub-doc
AI Summary	✓ Available	aiSummary: summary, keyInsights, suggestedNextAction, seriousnessScore, conversionProbability
Telecaller metrics	✓ Available	firstViewedAt, firstCallAt, totalCalls, totalActions, lastActionAt
Qualification answers	✓ Available	qualificationAnswers Map + qualificationProgress
Lead interests (structured)	✓ Available	interests sub-doc: courses, mode, location, placement, urgency, affordability, demoInterest
Lost reason tracking	✓ Available	lostReason, lostReasonCategory, lostAt, reEngagementDate
Eligibility	✓ Available	eligible / not_eligible / needs_review
Converted student link	✓ Available	convertedStudentId ref
Cost data	✓ Available	costData.costPerLead, campaignSpend, visible flag
Multi-tenancy (tenantId)	✓ Available	Required field with compound indexes
Duplicate phone check (DB)	⚠ Partial	Checked in Meta/WhatsApp webhooks only (last-10-digit regex). No unique DB index on {tenantId, phone}
Language preference (Telugu)	✗ Missing	No language field. WhatsApp messages are English only.

Feature / Field	Status	Notes
Demo booked date/time	✖ Missing	demoInterest: boolean exists but no demoBookedAt DateTime
Call SLA breach tracking	✖ Missing	No firstCallSLABreached or callSLABreachAt field

Supporting Models (all exist)

Model	File	Purpose
LeadStage	models/LeadStage.ts	Stage pipeline with SLA, automation triggers (sendWhatsApp, setFollowUp), Kanban config
LeadStageHistory	models/LeadStageHistory.ts	Immutable audit trail of stage transitions with entry/exit time and duration minutes
LeadScoringConfig	models/LeadScoringConfig.ts	Scoring rules, qualification rules, round-robin + rule-based auto-assignment config
LeadPriorityConfig	models/LeadPriorityConfig.ts	Rule-based priority overrides + eligibility rules
LeadFormConfig	models/LeadFormConfig.ts	Custom fields, per-field visibility, stats card config, table column config
QualificationQuestionConfig	models/QualificationQuestionConfig.ts	Question builder with categories, answer types, score impact, stage-based visibility

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Backend API Audit

All routes with RBAC, validation, pagination status

Lead Routes — server/src/routes/leadRoutes.ts

Route	Method	Purpose	Auth/RBAC
GET /leads	GET	Paginated list with filters (stage, source, assignedTo, search, priority, dateFrom, dateTo)	✓ Role-scoped: ALL/TEAM/OWN
GET /leads/:id	GET	Lead detail with full activities	✓ Scoped
POST /leads	POST	Create lead manually — calls scoreAndAssignLead	✓ create_leads
PUT /leads/:id	PUT	Full update	✓ edit_leads
DELETE /leads/:id	DELETE	Delete lead	✓ delete_leads
PATCH /leads/:id/quick-update	PATCH	Telecaller-friendly: stage + note + followup in one call	✓ edit_leads
PATCH /leads/:id/stage	PATCH	Change stage with history tracking	✓ edit_leads
POST /leads/:id/activities	POST	Add activity (call/note/email/whatsapp) + recording upload (100MB multer)	✓ edit_leads
POST /leads/:id/convert	POST	Convert lead to student account	✓ convert_leads
GET /leads/analytics	GET	Funnel, source performance, team analytics	✓ view_lead_analytics
GET /leads/manager-board	GET	Per-employee: leads, today followups, overdue, stage breakdown	✓ manage_leads
GET /leads/audit-logs	GET	All audit actions with pagination	✓ manage_leads
GET /leads/my-performance	GET	Self performance — followups, overdue, activity counts	✓ view_leads
GET /leads/export	GET	XLSX export with filter support	✓ export_leads
POST /leads/import	POST	Bulk import from XLSX/CSV	✓ manage_leads

Meta Ads, WhatsApp & AI Routes

Route Group	Routes	Status
Meta Lead Ads metaLeadAdsRoutes.ts	GET/POST /meta-leads/webhook — verify + receive Meta form submissions	✓ Working
WhatsApp whatsappRoutes.ts	verify webhook, receive messages, send manual, bulk cold leads, mark cold	✓ Working
Lead AI leadAIRoutes.ts	check status, generate summary (GPT-4), get summary, quick insights, bulk generate	✓ Working
Lead Stages leadStageRoutes.ts	CRUD + reorder + initialize defaults	✓ Working
Lead Scoring leadScoringRoutes.ts	config CRUD + team-members + rescore-all	✓ Working

Route Group	Routes	Status
Lead Priority	config + rule CRUD + thresholds + recalculate	✔ Working
leadPriorityRoutes.ts		

Missing APIs (Gap Table)

Missing API	File to Add	Priority
GET /leads/check-phone?phone=xxx — duplicate check before create	leadController.ts	High
GET /leads/overdue — leads where nextFollowUp < now and not closed	leadController.ts	High
POST /leads/:id/demo-book — book demo with date/time	New leadDemoController.ts	High
Follow-up reminder scheduler (node-cron)	New scheduler/reminders.ts	High
Daily summary email (cron at 8pm)	New scheduler/dailyReport.ts	High
GET /leads/aging?stageId=x&days=7 — leads stuck in stage	leadController.ts	Medium
POST /leads/:id/request-close — manager approval workflow	leadController.ts	Medium
WhatsApp conversation state persistence (replace in-memory Map)	whatsappWebhookController.ts	Critical
Multi-tenant Meta webhook routing (remove DEFAULT_TENANT_ID hardcoded)	metaLeadAdsController.ts	Medium

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Frontend UI Audit

client/src/pages — all lead-related pages, mobile support, fake-update prevention

Page / Folder	Status	Purpose & Notes
Leads/index.tsx	✓ Exists	Lead list table — filters (stage, source, priority, search, date range), configurable columns, pagination, import/export buttons
LeadDetail/index.tsx	✓ Exists	Full lead profile — activities timeline, AI insights panel, WhatsApp engagement, qualification answers, stage change, note/call logging. Sub-components: LeadDetailModern, LeadDetailV2
TelecallerConsole/index.tsx	✓ Exists	Telecaller-optimized view — lead queue (priority/followup sort), quick action panel (call/note/whatsapp), qualification form, AI insights, sales content library, lost reason modal
LeadManagerBoard/index.tsx	✓ Exists	Manager overview — per-employee lead counts, today followups, overdue, stage breakdown, expandable rows, reassign modal
LeadMyPerformance/index.tsx	✓ Exists	Self-performance — assigned, followups, overdue, activity counts by day/week/month, recent activity list, completion rate
LeadAnalytics/index.tsx	✓ Exists	Funnel analytics, source performance, team performance, bottlenecks, lost reasons, daily lead trend
LeadAuditLogs/index.tsx	⚠ Bug	Audit log viewer exists but calls <code>.ok</code> on parsed JSON response — <code>authenticatedFetch</code> returns JSON, not Response object. Page likely broken.
LeadScoringSettings/index.tsx	✓ Exists	Scoring rules, qualification criteria, assignment mode (none/round-robin/rule-based), team pool, re-score all
LeadPrioritySettings/index.tsx	✓ Exists	Priority rule config, threshold settings, bulk recalculate
LeadStages/index.tsx	✓ Exists	Stage manager — create, reorder, SLA config, automation triggers
LeadFormSettings/index.tsx	✓ Exists	Custom field builder, stats card config, table column visibility config
QualificationSettings/index.tsx	✓ Exists	Question builder — categories, answer types, score impact, stage visibility
FollowUpCalendar/index.tsx	🔥 Mock Data	Calendar view EXISTS but calls <code>getMockFollowUps()</code> — HARDCODED fake data, no real API connected
SalesContentLibrary/index.tsx	✓ Exists	Brochures, videos for telecallers to share during calls
GoogleSheetIntegration/index.tsx	✓ Exists	Google Sheets sync configuration
Marketing/index.tsx	✓ Exists	Campaign analytics, ad capture, competitor tracking
Mobile-optimized telecaller view	⚠ Untested	TelecallerConsole not verified for mobile breakpoints
Walk-in quick-entry form	⚠ Partial	Full lead form exists but no 10-second quick walk-in capture modal
Demo booking UI	✗ Missing	No date/time booking modal in LeadDetail
Telugu language toggle	✗ Missing	No language switcher anywhere in the app
Call recording playback	⚠ Partial	recordingUrl stored in activity — no <code><audio></code> player rendered in LeadDetail
Manager approval queue	✗ Missing	No approval page for lost/close stage requests

Page / Folder	Status	Purpose & Notes
Lead aging / stuck-lead board	 Missing	No table showing leads stuck in same stage > N days
Duplicate lead detection/merge UI	 Missing	No UI to detect or merge duplicate phone numbers
Real-time hot lead alerts	 Partial	Socket.io set up but no hot-lead-specific toast notification

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Sales Process Gap Analysis

End-to-end lead lifecycle — available / partial / missing

Process Step	Status	File	Notes
Meta Ad → Auto save to LMS	✓ Available	metaLeadAdsController.ts	Webhook receives, fetches from Graph API v19.0, creates lead
WhatsApp opt-in → Lead capture	✓ Available	whatsappWebhookController.ts	Creates lead immediately on first message, assigns stage
Walk-in lead capture	⚠ Partial	Leads/index.tsx	source='walkin' exists, but no dedicated fast-capture modal
Duplicate check by phone	⚠ Partial	metaLeadAdsController.ts, whatsappWebhookController.ts	Done in webhooks (last 10 digits). NOT enforced on manual create. No DB unique index.
Auto-assign to counselor	✓ Available	leadScoringService.ts	Round-robin and rule-based assignment on creation via scoreAndAssignLead()
Auto WhatsApp on lead creation	⚠ Partial	whatsappWebhookController.ts	Stage triggers have sendWhatsApp flag, but NOT auto-fired for Meta/manual creates
AI WhatsApp qualification flow	✓ Available	whatsappWebhookController.ts	4-step flow: initial → name → graduation year → course → qualified. 🚩 State in in-memory Map (lost on restart)
Lead scoring Hot/Warm/Cold	✓ Available	leadScoringService.ts	Rules engine recalculates on create and update
Follow-up reminders	⚠ Partial	Lead.nextFollowUp field	Date stored. Dashboard shows overdue count. No background job sends actual reminders.
Call recording upload	✓ Available	leadRoutes.ts (multer 100MB)	Upload works. recordingUrl stored. No playback UI.
Notes and status update	✓ Available	TelecallerConsole, LeadDetail	Both views support notes + stage changes with activity log
Demo booking	✗ Missing	—	interests.demoInterest: boolean only. No date/time booking. No calendar integration.
Admission conversion to student	✓ Available	POST /leads/:id/convert	Creates student account from lead with data transfer
Manager performance dashboard	✓ Available	LeadManagerBoard/index.tsx	Per-employee breakdown with expandable rows
Daily summary email report	⚠ Partial	LeadMyPerformance/index.tsx	Page exists for manual viewing only. No scheduled email delivery.
Telugu WhatsApp messages	✗ Missing	whatsappWebhookController.ts	All 4 qualification messages are English only. Zero Telugu support.

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AI Sales Automation Readiness

WhatsApp API, Meta webhook, Telugu support, OpenAI scoring

Infrastructure Status

- WhatsApp Business API (Meta Cloud) — connected via `WHATSAPP_ACCESS_TOKEN` + `WHATSAPP_PHONE_NUMBER_ID`
- Meta Lead Ads webhook — connected via `PAGE_ACCESS_TOKEN` + `DEFAULT_TENANT_ID`
- OpenAI GPT-4 Turbo — connected via `OPENAI_API_KEY` (leadAIService.ts), with rule-based fallback
- Socket.io — configured for tenant-scoped real-time events (tenant_{tenantId} room)

AI Feature	Status	Notes
WhatsApp API integration	✓ Connected	Meta Cloud API v18.0 via <code>WHATSAPP_ACCESS_TOKEN</code> . Bidirectional messages working.
Meta Lead Ads webhook	✓ Connected	<code>PAGE_ACCESS_TOKEN</code> + <code>DEFAULT_TENANT_ID</code> must be in .env
AI lead summary (GPT-4)	✓ Available	leadAIService.generateSummary() — prompt includes full lead + activity history. 24hr cache.
Rule-based quick insights	✓ Available	generateQuickInsights() — fallback when <code>OPENAI_API_KEY</code> not set
Auto lead scoring	✓ Available	LeadScoringConfig rules engine. scoreAndAssignLead() called on create.
Round-robin auto-assignment	✓ Available	LeadScoringConfig.roundRobinMembers[] + roundRobinIndex tracker
Rule-based auto-assignment	✓ Available	assignmentRules with conditions on lead fields
WhatsApp qualification flow	⚠ Partial	4-step conversation works. 🚨 State is in-memory Map — server restart destroys all active conversations.
WhatsApp conversation persistence	🚨 Critical Bug	<code>const conversationStates = new Map<string, LeadQualificationData>()</code> — in-memory only. Must move to MongoDB/Redis.
Reminder automation	✗ Missing	No cron/scheduler. No push notifications. No reminder emails or WhatsApp reminders.
Daily summary report (auto email)	✗ Missing	No scheduled job. No email automation for daily/weekly reports.
Missed follow-up alerts	⚠ Partial	Dashboard shows overdue count. No automated alert sent to manager/counselor.
Hot lead real-time alert	⚠ Partial	Socket.io configured. No specific 'lead:hot_created' event emitted.
Lead aging / stuck-stage detection	✗ Missing	No backend query or cron for leads stuck > X days in same stage.
AI-generated Telugu messages	✗ Missing	OpenAI prompts are English-only. No language parameter in leadAIService.ts.
Multi-tenant Meta webhook	✗ Missing	<code>DEFAULT_TENANT_ID</code> hardcoded in metaLeadAdsController.ts — not real multi-tenancy.

🚨 Critical Issue: WhatsApp Conversation State

- Current: `const conversationStates = new Map<string, LeadQualificationData>()` in whatsappWebhookController.ts
- Impact: Every server restart (deployment, crash, scaling) destroys all in-progress qualification conversations

- Fix: Create a `WhatsAppConversationState` MongoDB collection with phone, tenantId, step, collectedData, lastMessageAt, expiresAt (TTL index)

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Employee Control & Accountability

Tracking, fake update detection, SLA breach, manager approvals

Accountability Feature	Status	Notes
Lead assigned timestamp	✓ Available	assignment.assignedAt + previousAssignees history
First call time tracking	✓ Available	telecallerMetrics.firstCallAt recorded
Total call attempts	✓ Available	telecallerMetrics.totalCalls incremented per call activity
First view time (telecaller)	✓ Available	telecallerMetrics.firstViewedAt — when the counselor first opened the lead
Time to first action	⚠ Partial	telecallerMetrics.firstActionAt vs assignedAt can be computed but no SLA alert
Missed follow-up count	⚠ Partial	Overdue count shown in dashboard but not stored on User profile for history
Counselor-wise conversion rate	✓ Available	getManagerBoard API returns stage breakdown per employee
Activity audit trail	✓ Available	AuditLog collection records all mutations with IP address, userId, before/after values
Fake status update detection	⚠ Partial	AuditLog has all records + IP. No automated fraud scoring or pattern detection.
Manager approval for lost/close	✗ Missing	Anyone with edit_leads can close a lead. No approval workflow.
Call recording proof	⚠ Partial	Upload supported (multier 100MB), recordingUrl stored in activity. No playback UI in LeadDetail.
Daily telecaller report (automated)	✗ Missing	Manual page visit only. No scheduled daily productivity email.
SLA breach flagging	✗ Missing	LeadStage has SLA config (maxDurationHours, alertAfterHours) but no background job evaluates it
Re-assignment on inactivity	✗ Missing	No auto-reassign if counselor doesn't act within X hours

Fake Update Prevention — What's needed

- Require call recording upload before marking call as "connected" in TelecallerConsole
- Manager approval step before moving lead to "Lost" or "Closed" stages
- SLA cron: if firstCallAt is null and (now - assignedAt) > 2 hours → alert manager via socket/email
- Suspicious pattern detection: counselor marking >20 leads as "not_answered" in 1 hour → flag for review

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Database Gap Report

Missing fields, indexes, and new collections required

Collection	Existing	Missing / Add	Priority
Lead	phone (string, no unique)	Compound unique index: {tenantId, phone} with softDelete support	High
Lead	interests.demoInterest: boolean	demoBookedAt: Date, demoMeetingLink: string	High
Lead	telecallerMetrics (basic)	callSLABreached: boolean, callSLABreachAt: Date, firstCallSLAHours: number	High
Lead	—	languagePreference: 'english' 'telugu' 'hindi'	High
Lead	—	managerApprovalRequired: boolean, managerApprovedBy: ObjectId, managerApprovedAt: Date	Medium
Lead > activities	recordingUrl: string	recordingDurationSeconds: number, recordingFileSizeBytes: number	Low
WhatsAppConversationState	✗ Doesn't exist (in-memory Map)	New collection: phone, tenantId, step, collectedData (name/year/course), lastMessageAt, expiresAt (TTL 24hr)	Critical
FollowUpReminder	✗ Doesn't exist	New collection: leadId, assignedTo, scheduledAt, reminderType (email/whatsapp/sms), status (pending/sent/cancelled), sentAt	High
DailyPerformanceReport	✗ Doesn't exist	New collection: tenantId, date, perUser stats snapshot, totalLeads, conversions, callsMade, followUpsMissed	Medium
LeadStage	triggers.onEnter.sendWhatsApp	triggers.onEnter.whatsappTemplateBody (Telugu + English), triggers.onEnter.whatsappLanguage	Medium
LeadScoringConfig	roundRobinMembers[]	autoReassignAfterHours: number, autoReassignOnSLABreach: boolean	Medium
User	leadDataScope: string	telecallerLanguage: 'english' 'telugu', dailyCallTarget: number	Medium

Index Recommendation

- Lead: { tenantId: 1, phone: 1 } unique: true — prevents duplicate leads at DB level
- Lead: { tenantId: 1, nextFollowUp: 1, stageId: 1 } — powers overdue query efficiently
- WhatsAppConversationState: { expiresAt: 1 } expireAfterSeconds: 0 — TTL auto-cleanup
- FollowUpReminder: { scheduledAt: 1, status: 1 } — cron query performance

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API Gap Report

Required endpoints — existing / missing / changes needed

Required API	Status	File Path	Changes Needed	Priority
Duplicate phone check before create	✗ Missing	server/src/controllers/leadController.ts	Add <code>GET /leads/check-phone?phone=xxx</code> + add {tenantId, phone) unique index	High
Overdue leads list endpoint	✗ Missing	server/src/controllers/leadController.ts	Add filter: <code>nextFollowUp < now</code> AND stage not closed	High
Demo booking (date + time)	✗ Missing	New: server/src/controllers/leadDemoController.ts	POST /leads/:id/demo-book { scheduledAt, meetingLink, notes }	High
Follow-up reminder scheduler	✗ Missing	New: server/src/schedulers/reminderJob.ts	node-cron every 15min: find pending reminders, send WhatsApp/email	High
Daily summary email (cron)	✗ Missing	New: server/src/schedulers/dailyReportJob.ts	node-cron at 8pm: aggregate daily stats per counselor, send email	High
Auto WhatsApp on lead creation	⚠ Partial	server/src/controllers/leadController.ts	After createLead: call sendWhatsApp with welcome template if phone available	High
Hot lead real-time alert	⚠ Partial	server/src/controllers/leadController.ts	After score: if priority becomes 'hot', emit <code>lead:hot_created</code> socket event	High
WhatsApp state persistence	🔥 Critical	server/src/controllers/whatsappWebhookController.ts	Replace in-memory Map with WhatsAppConversationState MongoDB collection	Critical
Lead aging report	✗ Missing	server/src/controllers/leadController.ts	GET /leads/aging?stageId=x&days=7 — find leads stuck > N days	Medium
Manager approval for lost/close	✗ Missing	server/src/controllers/leadController.ts	POST /leads/:id/request-close + PATCH /leads/:id/approve-close	Medium
Fix LeadAuditLogs API call	✗ Bug	client/src/pages/LeadAuditLogs/index.tsx	Remove .ok check — authenticatedFetch returns parsed JSON not Response	Medium
Multi-tenant Meta webhook routing	✗ Missing	server/src/controllers/metaLeadAdsController.ts	Replace DEFAULT_TENANT_ID with per-tenant config lookup by page_id or form_id	Medium
Telugu AI prompt generation	✗ Missing	server/src/services/leadAIService.ts	Add language param; include "respond in Telugu" instruction in GPT-4 prompt	Medium
Meta lead calls scoreAndAssignLead	✗ Missing	server/src/controllers/metaLeadAdsController.ts	Call scoreAndAssignLead(newLead._id) after Meta lead creation (manual create does this, Meta doesn't)	High

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UI Gap Report

Missing screens, broken pages, mobile issues

Required Screen	Status	File Path	Changes Needed	Priority
Walk-in quick-entry form	 Partial	client/src/pages/Leads/index.tsx	Add "Walk-in" quick button → minimal modal (name + phone + course = 3 fields, autofills source='walkin')	High
Follow-up calendar — real API	 Mock Data	client/src/pages/FollowUpCalendar/index.tsx	Replace getMockFollowUps() with real <code>GET /leads?nextFollowUpFrom=X&nextFollowUpTo=Y</code> API call	High
Telugu language toggle	 Missing	New component: LanguageToggle	Add language switcher in header; save preference to user profile; translate WhatsApp templates	High
Demo booking UI	 Missing	New modal in client/src/pages/LeadDetail/	Date/time picker → POST <code>/leads/:id/demo-book</code> . Show booked demo in activities timeline.	High
Mobile-responsive telecaller view	 Untested	client/src/pages/TelecallerConsole/index.tsx	Test on 375px/768px. Fix stacked layout for call action panel and lead queue.	High
Hot lead real-time toast notification	 Partial	client/src/App.tsx or layout	Listen for <code>lead:hot_created</code> socket event → show toast with lead name and quick-open link	High
Call recording playback	 Partial	client/src/pages/LeadDetail/index.tsx	In activity timeline: if <code>activity.recordingUrl</code> → render <code><audio controls></code> player	Medium
LeadAuditLogs bug fix	 Bug	client/src/pages/LeadAuditLogs/index.tsx	Remove <code>.ok</code> check on <code>authenticatedFetch</code> result (it returns JSON, not Response object)	Medium
Manager approval queue	 Missing	New: client/src/pages/LeadApprovals/index.tsx	List leads with <code>managerApprovalRequired=true</code> . Approve/reject with reason.	Medium
Lead aging board	 Missing	New: client/src/pages/LeadAging/index.tsx	Table of leads stuck > N days per stage. Color-coded urgency rows.	Medium
Duplicate lead merge UI	 Missing	New: client/src/pages/LeadDuplicates/index.tsx	Group by last-10-digits of phone, show duplicates, "Merge" action.	Medium
Daily report dashboard (manager)	 Missing	New: client/src/pages/LeadDailyReport/index.tsx	Yesterday's performance per agent: calls made, followups completed, conversions.	Medium
Walk-in attendance quick log	 Missing	Leads/index.tsx or new page	Kiosk-style view for front desk: name + phone → auto create walkin lead	Low

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Final Recommendations

Strength assessment and phased implementation roadmap

What Is Already Strong — Don't rebuild, just extend

- Lead data model is **enterprise-grade** — more fields and sub-documents than most commercial CRMs like Zoho or Freshsales
- Meta Lead Ads webhook is **fully working** — receives, fetches from Graph API, creates lead automatically
- WhatsApp bidirectional bot with qualification flow (name → graduation year → course interest)
- AI summaries via OpenAI GPT-4 Turbo with rule-based fallback — already integrated
- Round-robin AND rule-based auto-assignment engine — already configured
- TelecallerConsole, ManagerBoard, MyPerformance, Analytics — 4 role-specific views ready
- Full audit trail via LeadStageHistory + AuditLog collection
- XLSX import/export, custom form fields, configurable table columns
- Role-based data scoping (ALL/TEAM/OWN) via leadScope middleware — clean and correct

What Is Weak — Must be fixed before going to production

- 🚨 **WhatsApp conversation state is in-memory** — every server restart wipes active bot conversations
- 🚨 **No automated follow-up reminders** — telecallers only see counts, no push/email/WhatsApp reminder is ever sent
- 🚨 **FollowUpCalendar is entirely mock data** — the calendar page shows hardcoded fake events
- 🚨 **No duplicate phone enforcement** — no DB unique index, no UI warning on manual create
- 🚨 **Zero Telugu support** — all WhatsApp bot messages are English only
- 🚨 **No demo booking feature** — critical gap in the admissions funnel
- 🚨 **Meta webhook single-tenant** — DEFAULT_TENANT_ID hardcoded, not true SaaS
- 🚨 **Meta lead creation skips scoreAndAssignLead()** — Meta leads not scored or assigned automatically

Implementation Roadmap

PHASE 1 — WEEK 1

Critical Fixes

- Persist WhatsApp state to MongoDB (WhatsAppConversationState collection)
- Connect FollowUpCalendar to real API
- Add phone unique index + duplicate check on manual create
- Auto-WhatsApp welcome message on lead creation
- Call scoreAndAssignLead after Meta lead creation
- node-cron follow-up reminder job (15min interval)
- Walk-in quick-entry modal (3-field form)

PHASE 2 — DAYS 8-15

Core Features

- Telugu WhatsApp message templates
- Demo booking modal + demoBookedAt field
- Daily summary email (node-cron 8pm)
- Manager approval workflow for lost/close
- Call recording <audio> player in LeadDetail
- Hot lead real-time socket toast
- Fix LeadAuditLogs .ok bug
- SLA breach flagging cron

PHASE 3 — LATER

Scale & Polish

- AI-generated Telugu follow-up messages
- Per-tenant Meta webhook routing
- Lead aging board + stuck-lead alerts
- Duplicate merge UI
- Mobile PWA for telecallers
- Google Sheets real-time sync
- Kanban drag-drop board view
- Advanced conversion analytics

Monthly Running Cost Estimate

Service	Usage (200 leads/day)	Estimated Cost/Month
WhatsApp Business API (Meta Cloud)	~6,000 conversations/month (1,000 free)	₹1,800 – ₹2,400
OpenAI GPT-4 Turbo	200 summaries × ~500 tokens = ~\$1/day	₹2,500 – ₹3,000

Service	Usage (200 leads/day)	Estimated Cost/Month
MongoDB Atlas (M10 or M20)	Current usage	₹1,500 – ₹2,000
Total Estimate	—	₹5,800 – ₹7,400/month

Summary Verdict

- The **backend is 75% complete** — schema and most APIs are production-ready
- The **frontend is 60% complete** — key views exist but FollowUpCalendar is broken and several screens are missing
- The **automation layer is 40% complete** — WhatsApp bot works but is fragile (in-memory state), no reminder cron, no daily emails
- **Estimated time to production-ready:** 15 working days (Phase 1 + Phase 2)
- **Do NOT rebuild** — the foundation is strong. Focus exclusively on the gaps listed above.